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FACULTAD DE CIENCIAS FÍSICAS Y MATEMÁTICAS

VALIDACIÓN DEL INSTRUMENTO DE BAJO COSTO PARA CARACTERIZACIÓN ATMOSFÉRICA DE CePIA

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Resumen

El vapor de agua es la molécula más importante en el proceso de absorción radiativa en el rango sub-mm/mm, por lo que su estudio en el testeo de sitios para instalar radiotelescopios es esencial. Normalmente se estudia con Radiómetros de Vapor de Agua, pero estos instrumentos son caros y complejos. Para alivianar este proceso, CePIA (UdeC) ha desarrollado un instrumento de Bajo Costo que potencialmente puede medir el Vapor de Agua Precipitable presente en la atmósfera. En este trabajo estudiamos un instrumento ubicado en GARS, Antártica y otro ubicado en APEX, Chile. Ambos tienen integrados varios sensores de radiación y termómetros infrarrojos, y logramos encontrar una relación exponencial entre la Temperatura Infrarroja del Cielo y el Vapor de Agua Precipitable. Utilizamos el satélite MERRA-2 de la NASA para comparar las mediciones del GARS, y el radiómetro APEX para el segundo instrumento con el fin de validar los datos.

Abstract

Water Vapor is the most important molecule in the process of radiative absorption in the sub-mm/mm range, so it's study in the site testing task for install radiotelescopes is essential. This is usually studied with Water Vapor Radiometers, but they are expensive and difficult. In order to alleviate this process, CePIA (UdeC) has developed a Low Cost instrument that can potentially measure the Precipitable Water Vapor present in the atmosphere. One instrument located at GARS, Antarctic and other one located at APEX, Chile are studied in this research. Both has integrated several radiation sensors and infrared thermometers, and we were able to find an exponential relationship between the Infrared Sky Temperature and the Precipitable Water Vapor. We use NASA's MERRA-2 sattellite to compare the GARS measurements, and the APEX Radiometer for the second instrument in order to validate the data.

Keywords – PWV - Site Testing - Atmospheric Characterization - Astroclimatology

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Chapter 1

Context

1.1 Introduction

Radiotelescopes are located at high altitudes to avoid the absorption of radiation produced by atmospheric water vapor ([Smith \(1985\)](#)). Water vapor is the main source of opacity in the atmosphere at mm/sub-mm wavelengths, therefore, its concentration is considered the most relevant criteria for evaluating a potential site for a radiotelescope ([Li et al. \(2020\)](#)). Typically, there is an exponential decrease of the water vapor density with altitude up to 2km ([Starr and White \(1955\)](#)), and this is the reason why these instruments are located at high and dry zones.

Unlike other atmospheric gases, the water vapor concentration varies over time, altitude and geographical location, and is directly related to the weather conditions ([Querel and Kerber \(2014\)](#); [Smette, A. et al. \(2015\)](#)). Therefore, determining whether a site is optimal for the installation of a radiotelescope is crucial a long-term and continuous study of the precipitable water vapor ([Valeria \(2023\)](#)). This study is typically made with Water Vapor Radiometers that measures the radiation emitted by the water vapor molecule. The more the radiation measured, the more the ammount of water vapor, and therefore the absorption is greater.

1.2 Precipitable Water Vapor

Water vapor is a molecule composed of two hydrogen atoms and one oxygen atom. This molecule has a small moment of inertia in rotation, leading to a combination of vibrational-rotational spectra in the vapor, which contains many absorption lines (Engelen and Stephens (1999)). These vibrations occur in three modes: symmetric stretching, bending, and asymmetric stretching (Braun and Smirnov (1993)), and the energy of these vibrations is obtained by absorbing radiation at specific frequencies: 22 GHz, 183 GHz and 448 GHz, among others. Figure 1.2.1 shows the emission lines of water vapor at these frequencies, among other molecules emissions lines, like O₂.

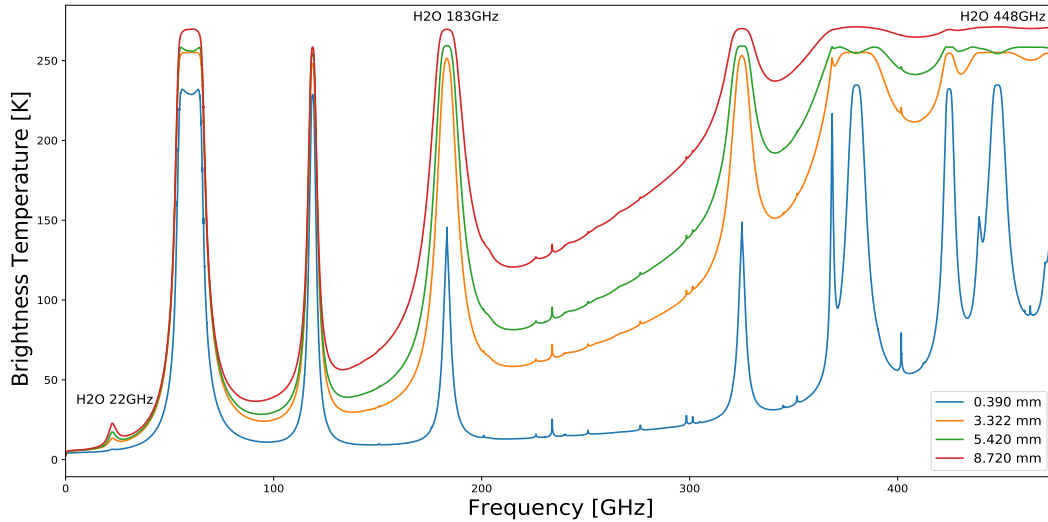


Figure 1.2.1: Simulated spectra of the atmosphere showing some of the water vapor emission lines using the AM software (Paine (2024)). Each color represents a location in the world with different amount of precipitable water vapor.

Water vapor plays a crucial role in the processes of radiative absorption and emission in the atmosphere: relatively small amounts of water vapor can cause significant variations in climate (Mohanakumar (2008)), therefore, any disturbance in its abundance can have diverse impacts. This makes water vapor one of the most dominant species in terms of absorption in the Earth's atmosphere across the microwave spectrum (Valeria (2023)).

Precipitable Water Vapor (PWV) is the total amount of water vapor that would fall to the ground if all the liquid water in the atmospheric column were condensed and precipitated (Marín et al. (2015)). The PWV is typically used to diagnose

atmospheric humidity over a location and is usually measured in millimeters.

As water vapor is one of the most important molecules in the absorption of microwave radiation, it is a critical factor in the site testing task when choosing a location to install a radiotelescope.

1.3 The Event Horizon Telescope

The Event Horizon Telescope (EHT) is an international scientific collaboration aimed at studying Super Massive Black Holes (SMBHs). It employs interferometry, a technique in radio astronomy that enables the observation of details in objects that cannot be resolved even by the largest individual telescopes. This method combines the light captured by several telescopes to create a more detailed image of the object than would be possible with each individual telescope. It effectively creates a virtual telescope with improved angular resolution, following the equation $\theta \propto \lambda/D$, where λ is the observed wavelength and D is the diameter of the telescope. As the diameter increases, the angular resolution decreases, allowing for more defined details of the objects.

Until 2017, the EHT combines eight arrays of mm/sub-mm telescopes: SMT, APEX, IRAM-30m, JCMT, LMT, SMA, ALMA, and SPT. This configuration (Figure 1.3.1) allows the EHT to create an Earth-sized virtual telescope and achieve an angular resolution of less than $20\mu\text{as}$ (Akiyama et al. (2019)). The EHT is responsible for the images of the SMBH in the elliptical galaxy M87 and Sgr A*, the SMBH of our own galaxy, both obtained at 230 GHz (~ 1.3 mm).

The next-generation Event Horizon Telescope (ngEHT) aims to capture even sharper images of SMBHs. It will add approximately 10 new antennas, providing instrumental upgrades to the existing antennas, and increase sensitivity, frequency, and uv-plane coverage (Roelofs et al. (2023)). For this purpose, several studies have evaluated candidate sites for new arrays or single-radio telescopes to be part of the ngEHT, or for atmospheric characterization (Suen et al. (2014); Raymond et al. (2021); Rioja et al. (2023)).

For example, in Figure 1.3.1, Raymond et al. (2021) proposes more than 40 potential sites for the installation of arrays or single-radio telescopes, based on the work of Suen et al. (2014), which investigates the amount of precipitable water



Figure 1.3.1: Configuration of the eight EHT telescopes around the world.

vapor present in different parts of the world.

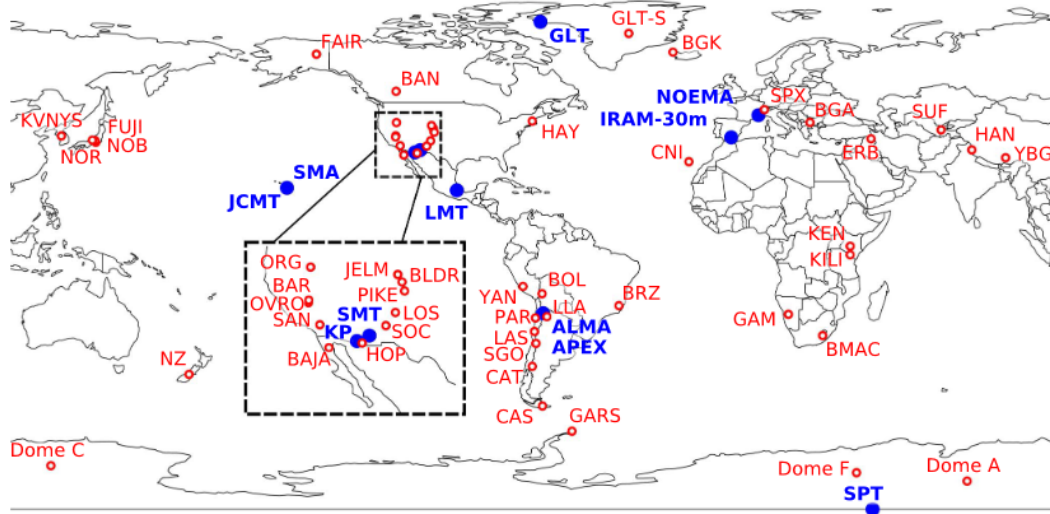


Figure 1.3.2: Map of EHT sites until 2021 (blue dots) and 45 potential new sites for the ngEHT (red dots). Image from [Raymond et al. \(2021\)](#).

1.4 Motivation

For the atmospheric water vapor characterization, the most common tool are the Water Vapor Radiometers. These instruments measured the electromagnetic power radiated from an object in the microwave region ([Peña \(2023\)](#)). The thermally radiated energy is proportional to the physical temperature, so the

radiometer provides a measure of the object’s temperature through the Rayleigh-Jeans approximation (Kämpfer (2012)). The water vapor in the atmosphere emits microwave radiation at specific frequencies (see Section 1.2), and the amount of radiation emitted is influenced by the temperature and concentration of water vapor (Schneider et al. (1999)). This is the radiation that the Water Vapor Radiometers measure.

Water vapor radiometers are designed to observe at specific frequencies, with the most common being the 22 GHz and the 183 GHz Radiometers. However, the main problem with these instruments is that they are expensive and complex (Askne and Westwater (1986); Rose et al. (2005)), ranging from a few million dollars to several hundred million dollars. This high cost has prompted several studies to explore low-cost instruments that can replace these radiometers for site-testing purposes (Daniels (1998); Stephan and Pearce (2002); Britcliffe et al. (2012); Karkare et al. (2014); Hernandez and Saini (2019)).

1.5 This work

The Centro Para la Instrumentación Astronómica (CePIA) at Universidad de Concepción, in collaboration with the Millennial Nucleous TITANS, has developed an low-cost instrument that could work as a replacement for the Water Vapor Radiometers in the site-testing task. This instrument will potentially provide PWV measurements by measuring the infrared clear-sky temperature.

The main aim of this instrument is to provide atmospheric information, giving insights of water vapor concentration. According to Figure 1.3.2, one of the possible sites for the installation of new arrays is the GARS station in Antarctic. Therefore, the first low-cost instrument was located there from February to October, 2023.

In addition to this, a new and improved instrument was installed at the APEX location, on the Chajnantor Plateau, since October 2024.

This work is the study of this instrument and the validation of its data.

1.5.1 Related Works

The interest in Antarctica for astronomical studies is well-established; the Antarctic Plateau offers a combination of cold, dry, and tenuous air, along with relatively

stable weather conditions, making it ideal for astronomical observations ([Burton \(1995\)](#)). Studies such as [Storey et al. \(2000\)](#); [Storey et al. \(2002\)](#); [Lawrence et al. \(2006\)](#); and [Falvey and Rojo \(2016\)](#) all suggest that Antarctica is a viable location for astronomical research.

In the use of infrared sky measurements, [Maghrabi et al. \(2009\)](#) and [Maghrabi and Clay \(2010\)](#) used infrared thermometers to measure the clear-sky temperature and found both linear and exponential relationships, supporting the objective of the CePIA low-cost instrument.

Similar to others, [Sharma \(2016\)](#) used a thermopile for cloud detection for an optical telescope, and [Rafiei \(2006\)](#) developed a low-cost instrument for measuring astronomical seeing. Moreover, [Ziad et al. \(2022\)](#) developed a ground-based instrument to measure cloud cover and atmospheric turbulence.

Although these are all similar in the sense of low-cost site-testing instruments, none can replace water vapor radiometers in site-testing tasks like the CePIA monitor intends to do.

Chapter 2

Theoretical Background

2.1 Low-Cost Site Testing Instrument

The Low-Cost Site Testing Instrument (from now on, LCST), was first developed in 2020 at CePIA laboratory. It is based on a Raspberry Pi and photodiodes, covering from the ultraviolet to the near-infrared spectral range.

In February 2023, one lcst station was installed on the roof of the GARS base, and it operated until October 2023.

In October 2024, a second lcst station was installed on the roof of a control room in Chajnantor Plateau, near the APEX telescope and its radiometers. Until this report, it is still operating.

A detailed study of the instrument will be presented in the following sections.

2.1.1 Sensors

The instrument consists mainly of four sensors: three related to electromagnetic radiation and one temperature sensor. They all operate within temperature ranges from -40°C to 85°C , feature analog-to-digital converters, and communicate via the I2C serial protocol. The radiation sensors work through the photoelectric effect, and the temperature sensor relies on detecting infrared radiation emitted by objects. Additionally, the instrument is capable of taking data approximately every 30 seconds.

2.1.2 VEML 7700

This is an AMBIENT LIGHT SENSOR (Electronics (2024)). It operates through a photodiode and has two output channels: one with a spectral response similar to the human eye in the 450-650 nm range, and a second one designated as "white light" in the 400-1050 nm range. Each channel provides an integral response over its spectral range.

It outputs values are in *lux*, which measures the amount of light on a surface per unit area integrally; it delivers a single value for its entire bandwidth with a resolution of 0.0042 [lx].

In figure 2.1.1a we can see the responsivity of the light sensor, and in figure 2.1.1b we can see the reponsivity of the white channel.

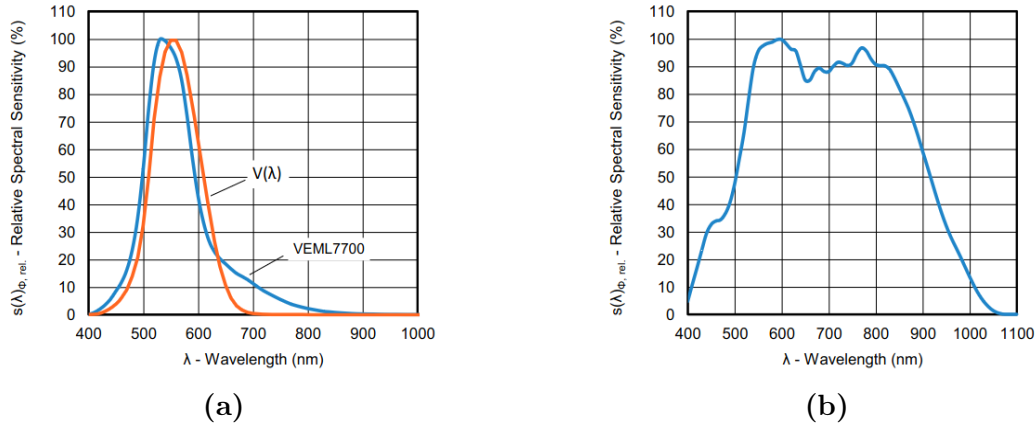


Figure 2.1.1: Spectral response of both channels of the VEML77000.

2.1.3 VEML 6075

This is an ULTRAVIOLET RADIATION SENSOR (Vishay Intertechnology, Inc. (2023)). It works through a photodiode and has two channels: one UVA channel in the 320-400 nm range and one UVB channel in the 280-320 nm range. It provides a spectral response in its range.

It outputs values are in $[\mu W/cm^2]$, indicating the power flux density, or how much electromagnetic power is passing through a given area. The UVA channel has a responsivity of 0.93 $[ct/\mu W/cm^2]$, and the UVB channel has a responsivity of 2.1 $[ct/\mu W/cm^2]$.

In figure 2.1.2, we can see the responsivity of the sensor for both channels. Note that there are compensation channels for both UVA and UVB to correct the radiation values.

It is important to mention that this sensor was not calibrated with black body radiation, so the raw data obtained is only proportional to $\mu W/cm^2$.

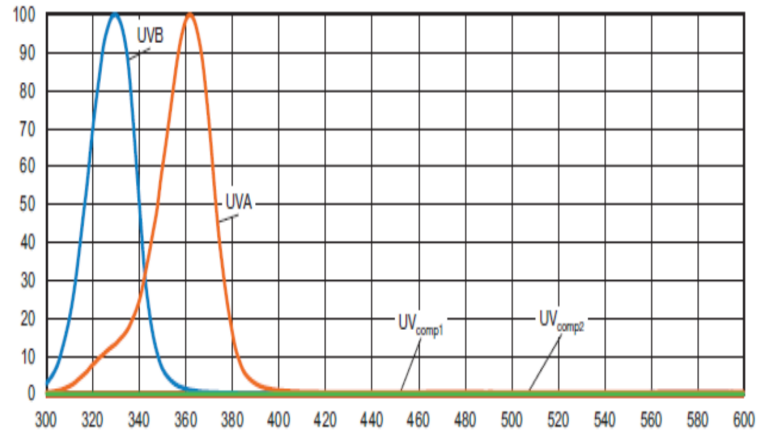


Figure 2.1.2: Spectral response of both channels of the VEML6075.

2.1.4 AS7265X

This is a TRIAD OF SPECTROSCOPES ([ams OSRAM Group \(2023\)](#)). It consists of three spectral sensors:

- AS72651: in the 610-860nm range (V-NIR).
- AS72652: in the 560-940nm range (V-NIR).
- AS72653: in the 410-535nm range (NUV-V).

Each has 6 channels and a bandwidth of approximately 20 nm, covering a total of 18 channels from the near-UV to the near-infrared. It provides a spectral response in its range. It outputs values are in $[\mu W/cm^2]$, indicating the power flux density.

In figure 2.1.3, we can see the responsivity of the 18 channels of the sensor.

Similar to the VEML6075, this sensor was not calibrated with black body radiation, so the raw data obtained is only proportional to $\mu W/cm^2$.

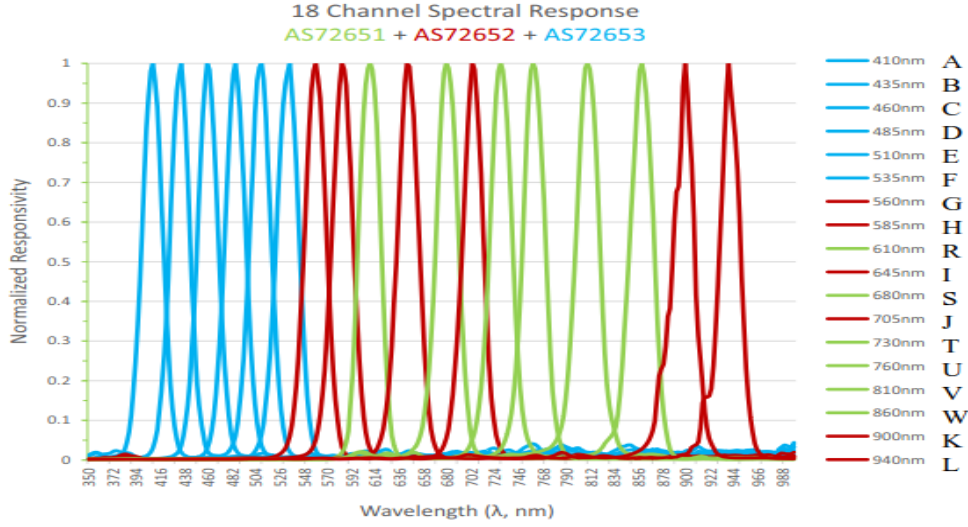


Figure 2.1.3: Spectral response for all the 18 channels of the AS7265X. Each color corresponds to a different spectroscope, as labeled in the figure.

2.1.5 MLX 90614

This is an INFRARED THERMOMETER (Melexis (2023)). It operates through a thermopile, obtaining measurements of objects without contact. It has two channels: one for the object temperature (ranging from -70°C to $+380^{\circ}\text{C}$) and the other for the sensor temperature (ranging from -40°C to $+125^{\circ}\text{C}$).

It outputs temperature values in Kelvin with a resolution of approximately 0.02°C . Additionally, it has a wavelength passband filter for the optical and NIR range, operating from 5.5 to $14\ \mu\text{m}$.

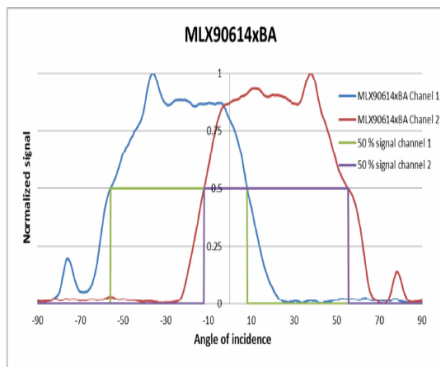


Figure 30: Typical FOV of MLX90614xBA

(a)

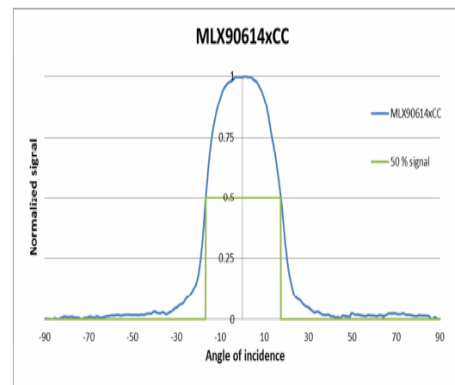


Figure 32: Typical FOV of MLX90614xCC

(b)

Figure 2.1.4: Typical field of view of the MLX90614 sensors. In the left panel is the BBA model and in the right panel is the BCC model.

The lcst uses two different models of this sensor: MLX90614-BBA (Fig 2.1.4a), which has a dual field of view, and MLX90614-BCC (Fig 2.1.4b), which includes temperature compensation. The typical field of view for both BBA and BCC sensors can be seen in figure 2.1.4.

This sensor was calibrated with an object of emissivity 1.0 (a black body), and the emissivity can be adjusted between 0.1 and 1.0.

2.1.6 BME280

This is an AMBIENTAL SENSOR (Bosch Sensortec (2023)). Its purpose is to measure the internal conditions of the instrument's housing. It has sensors for temperature, pressure and humidity.

The temperature sensor works in a range between -40°C to $+85^{\circ}\text{C}$. The pressure sensor measures between 300 and 1100 hPa. The humidity sensor provides values for relative humidity, between 0% and 100%.

In the lcst located at APEX, this sensor was installed in the exterior of the instrument, to monitoring the external conditions of ambient temperature, pressure and humidity.

2.1.7 DS3231

This is a INTERNAL CLOCK (Maxim Integrated (2023)). It features an internal crystal oscillator, which remains stable despite temperature changes, resulting in an error of less than 1 minute per year.

2.2 Design and configuration

All sensors are incorporated in a watertight enclosure. To hold the components inside the box, a structure was designed and fabricated using a 3D filament printer. Since the instrument uses a Raspberry Pi, control is managed by several Python scripts. These scripts allow for the automatic adjustment of both gain and integration time as needed. The instrument aims to achieve greater sensitivity under low brightness conditions and to avoid saturation when brightness is higher.

The latest version of the instrument is in Figure 2.2.1.



Figure 2.2.1: The latest version of the Low-Cost instrument, located at Chajnantor Plateau near the APEX radiometers.

2.3 MERRA2

The Modern-Era Retrospective Analysis for Research and Applications version 2 (MERRA-2) (Gelaro et al. (2017)) is NASA's latest atmospheric reanalysis, integrating observations from the satellite era with the Goddard Earth Observing System (GEOS) atmospheric data assimilation system.

MERRA-2 data is stored on a global grid with a horizontal resolution of 0.625° longitude by 0.5° latitude, comprising 580 longitude points and 360 latitude points. The dataset provides vertical information through several formats: *Horizontal-only*, which lacks vertical details; *Pressure-grid*, interpolated to 42 fixed atmospheric layers; *Model-level*, interpolated to 72 surface-dependent levels; and *Model-edge*, which calculates upper layer edges using pressure thickness.

Each type of file within MERRA-2 contains specific attributes such as specific humidity and ozone, among others, essential for atmospheric and climate research.

In this study, we will use an interpolation of the MERRA2 grid to find the PWV over a specific location: the GARS base in the Antarctic plain. This is based on the study of Valeria (2023), where, from MERRA2, we obtain the specific humidity at GARS, Q_v , and then we convert it to H_2O in volume mixing ratio, that defines the amount of water vapor present, through the following equation:

$$H_2O[vmr] = \left(\frac{m_{dry}}{m_{H_2O}} \right) \cdot \left(\frac{Qv}{1 - Qv} \right) \quad (2.3.1)$$

where H_2O represent the PWV, $m_{dry} = 28.964$ is the dry air molecular mass, $m_{H_2O} = 18.015$ is the water vapor molecular mass, and Qv is the specific humidity, defined as the ratio of water vapor mass per mass unit of total air.

2.4 APEX meteorological station

The data obtained by the Meteorological Station of the Atacama Pathfinder Experiment (APEX) ([Observatory \(2021\)](#)) is available in this [website¹](#). The dataset covers from January 2006 to the date, and it is being updated in real time. It delivers data every 1 minute of pwv, dew point, humidity, pressure, ambient temperature, wind direction and wind speed. Also, we can set some parameters like the azimuth and elevation of the telescope, and specific frequencies of the radiometer measurements.

In this study, we will use this data to validate and compare the lcst measurements. In particular, we are interested in the pwv, ambient temperature, humidity and pressure measurements of the APEX meteorological station.

¹http://archive.eso.org/wdb/wdb/eso/meteo_apex/form

Chapter 3

This Work

3.1 General Objective

The main objective of this study is to validate the data obtained by the Low-Cost Site Testing instrumentd for atmospheric characterization developed by CePIA.

3.2 Specific Objectives

- Reduce the data: the measured data comes in different formats and every ~ 30 seconds, independent of the climate, and we need to establish the useful data: the clear sky.
- Perform comparative analysis with radiometers or sattelites: to know if the predicted PWV is a good approximation, we need to compare it with known PWV. For this we use MERRA2 for the GARS measurements, and we use the APEX radiometer for the APEX-instrument.
- Perform an interpolation of MERRA2: to obtain correct pwv at the GARS location.
- Obtain data from radiometers: pwv from the APEX meteorological station.
- Find mathematical relationships between the PWV and the infrared sky temperature: so we can predict the pwv from the low-cost instrument sensors.

3.3 Ansatz: a first approach

We can relate the Infrared Sky Temperature with the PWV through two approaches:

- Linear relationship: following the equation 3.3.1:

$$PWV = a \cdot T_{sky} + b \cdot T_{amb} + c \quad (3.3.1)$$

where a, b and c are free parameters, T_{sky} is the infrared sky temperature, and T_{amb} is the ambient temperature.

- Exponential relationship: following the equation 3.3.2:

$$PWV = a \cdot \exp\left(\frac{T_{sky} + b}{c}\right) + d \quad (3.3.2)$$

where a, b, c and d are free parameters, and T_{sky} is the infrared sky temperature.

3.4 Limitations

One of the main limitation is the climate, specifically the clouds. The clouds affects all the sensors, in particular, the infrared thermometers by overestimating its measurements. This leads to wrong data that will affect the comparative analysis.

Due to this problem, the main focus of this work will be done with the lcst monitor located at APEX. The climate there is stable and has over 300 clear nights per year, unlike the Antarctic which most of the days are cloudys.

Chapter 4

Methodology

For this study, we will validate the CePIA low-cost site testing instrument by comparing it with the APEX radiometer data from its meteorological station. This validation will be done by finding a mathematical relationship between the infrared sky temperature and the pwv.

After the validation process, we will perform a cross-validation method in order to verify the performance of this mathematical relationship: we will use 75% of the data to find a relationship between these variables, and the 25% restant to see the performance of this relationship. We will extract statistical metrics and correlation coefficients.

4.1 GARS LCST

4.1.1 Data reduction

The first version of the Low-Cost Site Testing instrument was located at GARS, Antarctic, and it delivered data every 30 seconds, from February 01 to October 25, 2023. Despite this huge amount of data, we need to establish a point of comparison, in order to validate the data, so we use two tools: the MERRA2 interpolation, to obtain pwv, and the Meteorological Station from the [Dirección Meteorológica de Chile, DGAC](#)¹. They have a station located near the lcst, and it delivers data of ambient temperature, pressure and humidity.

¹<https://www.meteochile.gob.cl/PortalDMC-web/index.xhtml>

According to the equation 3.3.1, we need both the sky temperature and the ambient temperature, but as we said in section 2.1.6, the instrument located at GARS doesn't have an ambient temperature to monitoring the exterior, only the internal conditions, so we use the MeteoCHILE data to supply this.

Now, both MeteoCHILE and MERRA2 gives less data than the lcst: only one value every 3 hours. This leads us with a maximum of 8 points per day, in the following hours:

MERRA-2	01:30	04:30	07:30	10:30	13:30	16:30	18:30	22:30
MeteoCHILE	00:00	03:00	06:00	09:00	12:00	15:00	18:00	21:00

Table 4.1.1: Hours at which MERRA-2 and MeteoCHILE delivers data.

With this, we need to do two approaches: the first is to interpolate MERRA-2 data to temporarily match MeteoCHILE data, and secondly to match the lcst data with this data. To do the latter, we establish temporal window of ± 1.5 hours of the available data, centered at the corresponding hour, and then we average all the points.

But to establish which points are going to be average, we need to fulfill the main condition: a clear sky. The only approach we have to this, is a cloud-coverage measurement of MeteoCHILE in oktas. This measure unit is shown in Figure 4.1.1. To establish the useful data, we consider only the days that have less than 2 oktas, considering the field of view of the mlx sensors.

This restrictions leaves us with a total of 50 useful days along the 9 months considering the three sensors. This does not mean that each day was fully a clear day, but at least one data-point was taken with clear sky conditions. This is described in table 4.1.2, where in each month we numbered the amount of data-points.

Sensor	Feb	March	April	May	June	July	August	Sept	Oct	Total
BCC	24	1	15	0	35	34	14	0	0	123
BBA1	33	18	23	3	0	31	15	0	0	123
BBA2	0	0	0	3	0	32	15	0	0	50

Table 4.1.2: Available sky temperature data per month.

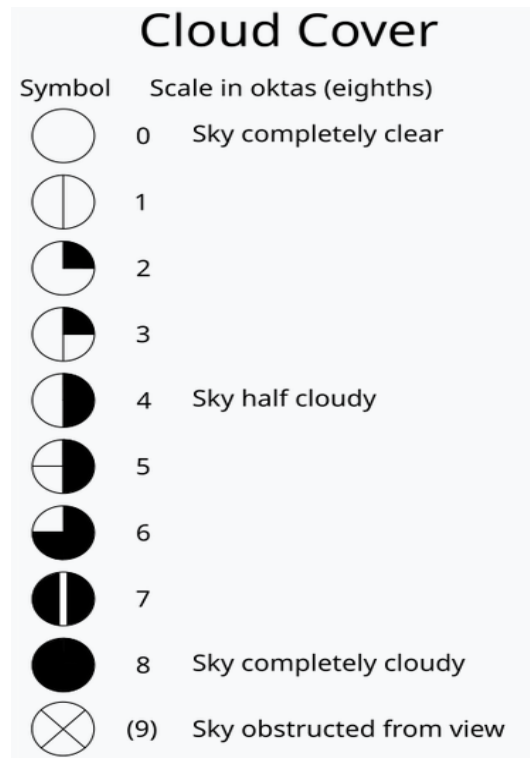


Figure 4.1.1: A representation of the cloud coverage using oktas as a measurement unit.

4.2 APEX LCST

4.2.1 Data reduction

The latest version of the Low-Cost Site Testing instrument is located at the Chajnantor Plateau, in the APEX location. It delivers data every 30 seconds, since October 15, 2024, to the date of this study. To establish a comparison point to validate the data, we will use the data from the [APEX radiometer²](#).

This version of the LCST has the BME-280 sensor in the outside, so we can monitor the ambient temperature, the specific humidity and the pressure conditions.

The radiometer from APEX delivers data each minute, so we can have 1440 points per day, so we will only have to match temporarily these two instruments. To do this, any lcst measurements made in the same minute will be averaged.

Also, the weather conditions are stable, with many days without clouds, so we can have a lot of reliable data, and we can obtain more accurate statistics.

²http://archive.eso.org/wdb/wdb/eso/meteo_apex/form

The lcast is located at the roof of a control room in APEX, pointing directly to the zenith. For this reason, when the sun rises and gets in the way of the sensors FoV, the temperature increases. As the sensors are sensitive to this increase due to the sun, we won't consider this measurements in the fit. For the BCC sensor, this contribution due to the sun is approximately between the 13:00 and the 18:30 [UTC]. For the BBA sensor, this contribution is between the 11:00 and the 22:30 [UTC] approx. This difference is due to the FoVs of each sensor, 35° and 75°, respectively.

The days in which we are going to extract statistical metrics are listed in table 4.2.1:

Month								Total
October	17	19	20	24	31			7200
November	01	02	03	04	06	07	08	10080

Table 4.2.1: Days used to find the mathematical relationships.

4.3 Statistical metrics

As we are making predictions of one parameter, which we are going to compare with the real value, we expect them to correlate in a 1:1 ratio. In order to quantify how good are the predictions, we will use several statistical metrics:

- MSE: it is the mean squared error, defined as:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (4.3.1)$$

where n is the total data points, y_i is the true value and $\hat{y}_i$ is the predicted value. The MSE gives the average squared difference between predicted and true values.

- R^2 score: it is a coefficient of determination, defined as:

$$R^2 = 1 - \frac{SS_{residual}}{SS_{total}} = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (4.3.2)$$

where $SS_{residual}$ measures the total squared differences between the true value and the predicted value, and SS_{total} measures the total squared differences between the true value and the mean of the true values.

- r : it is the Pearson Correlation Coefficient, defined as:

$$r = \frac{\sum_{i=1}^n (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_{i=1}^n (y_i - \bar{y})^2 \sum_{i=1}^n (\hat{y}_i - \bar{\hat{y}})^2}} \quad (4.3.3)$$

where $\hat{y}$ is the mean of the true values and $\bar{\hat{y}}$ is the mean of the predicted values. The numerator measured the covariance of the true and predicted values, and the denominator normalizes by the product of the standard deviations of the true and predicted values.

In general, R^2 gives insights on how well the predictions fits the true data, MSE measured the prediction accuracy, and the Pearson coefficient focuses on the strength of the linear relationship.

4.4 The Clausius-Clapeyron Equation

In addition to the equations mentioned in Section 3.3, we also found a new relationship to calculate the PWV: a modified Clausius-Clapeyron equation:

$$PWV = 0.493 \cdot e \cdot \frac{\left(1 + 1.61 \cdot \frac{e}{p}\right)}{T} \quad (4.4.1)$$

$$e = \frac{RH \cdot e_s}{100} \quad (4.4.2)$$

$$e_s = 6.11 \exp\left(\frac{17.27 \cdot T}{T + 237.3}\right) \quad (4.4.3)$$

where T is the ambient temperature in $^{\circ}\text{C}$, RH is the specific humidity in [%] and p is the pressure in [hPa]. The original Clausius-Clapeyron equation is the e_s term, that describes how the vapor pressure changes with temperature. Then we assume hydrostatic equilibrium and an ideal gas approximation to derive the PWV.

All of this measurements are provided by the BME280 sensor, so we focus on calibrating this sensor measurements as well.

Chapter 5

Results

Here we present the results of this work. Each section is for each LCST instrument, and then we divided it into linear, exponential or Clausius-Clapeyron equations, as needed.

This results are made by comparing the lcst measurements with MERRA-2 interpolated pwv, and with the APEX radiometer. Here we consider MERRA-2 and the radiometer as true values, so we create the mathematical relationships by fitting our measurements to this instruments.

5.1 GARS, Antarctic

5.1.1 Mathematical Relationships

According to the studies of [Maghrabi et al. \(2009\)](#) and [Maghrabi and Clay \(2010\)](#), we have the two equations described in [Section 3.3](#), but in a different way:

$$T_{sky} = a \cdot T_{amb} + b \cdot PWV + c \quad (5.1.1)$$

$$PWV = a \cdot \exp\left(\frac{T_{sky}}{b}\right) + c \quad (5.1.2)$$

5.1.1.1 Linear

By considering the total points of each sensor, described in [table 4.1.2](#), we were able to adjust the linear curve as follows:

- For the BCC sensor:

$$T_{sky} = 0.19 \cdot T_{amb} + 0.81 \cdot PWV_{MERRA-2} - 28.25 \quad (5.1.3)$$

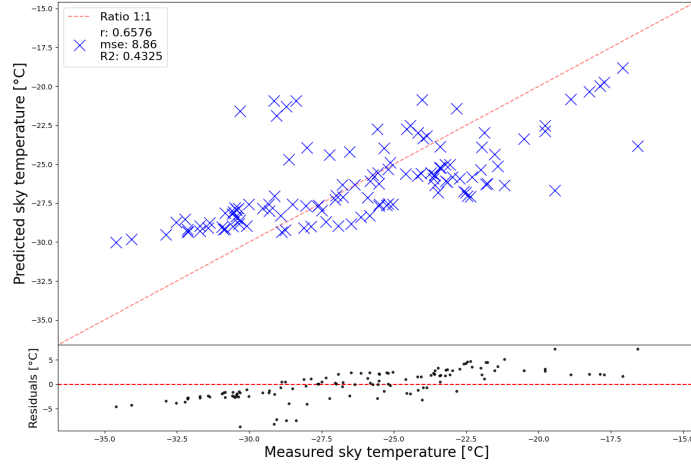


Figure 5.1.1: Comparison of the measured sky temperature versus the predicted sky temperature. The red line is a ratio 1:1. This prediction follows the equation 5.1.3, and has mse=8.85, r=0.65 and R2=0.43.

- For the BBA1 sensor:

$$T_{sky} = 0.23 \cdot T_{amb} + 1.97 \cdot PWV_{MERRA-2} - 41.53 \quad (5.1.4)$$

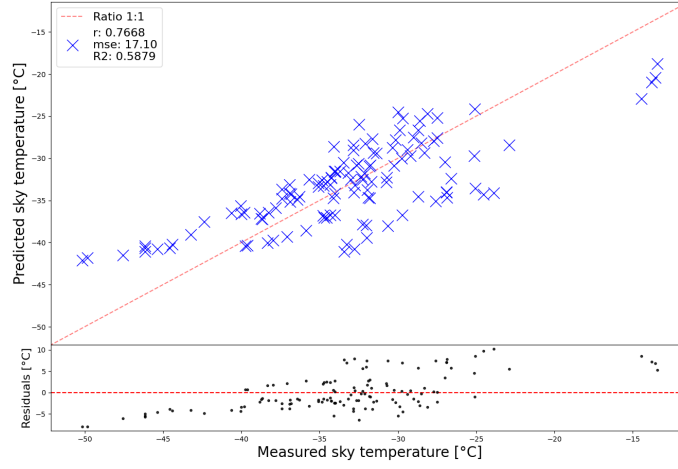


Figure 5.1.2: Comparison of the measured sky temperature versus the predicted sky temperature. The red line is a ratio 1:1. This prediction follows the equation 5.1.4, and has mse=17.09, r=0.76 and R2=0.58.

- For the BBA2 sensor:

$$T_{sky} = 1.33 \cdot T_{amb} - 0.53 \cdot PWV_{MERRA-2} - 16.90 \quad (5.1.5)$$

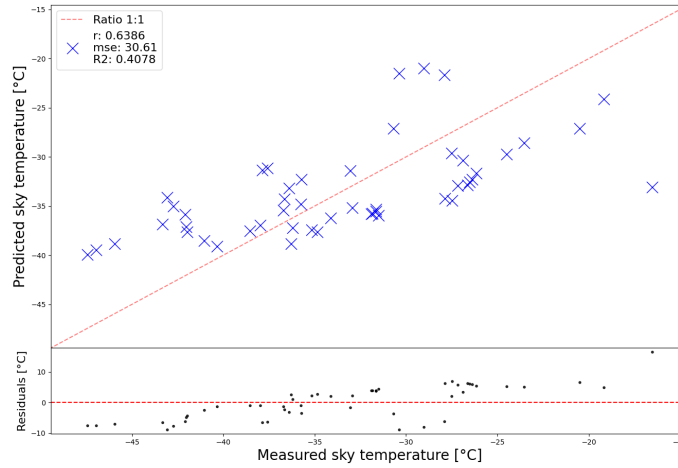


Figure 5.1.3: Comparison of the measured sky temperature versus the predicted sky temperature. The red line is a ratio 1:1. This prediction follows the equation 5.1.5, and has mse=30.61, r=0.63 and R2=0.40.

To make the fits, we use the CURVE FIT function from Scipy, considering a , b and

c as free-parameters to adjust to our data. The aim was to predict the pwv by inverting this equations, but this yields in nothing. The correct inversion of the equations are as follow:

$$PWV_{bcc} = 1.23T_{sky} - 0.23T_{amb} + 34.87$$

$$PWV_{bba1} = 0.50T_{sky} - 0.11T_{amb} + 21.08$$

$$PWV_{bba2} = -1.88T_{sky} + 2.50T_{amb} - 31.88$$

However, this does not yield in good results, giving poor metrics, specifically with the R2 score, which is < 0 in all cases.

Despite this, the sky predictions has no higher metrics either. We strongly believe that this is due to two things: the poor data obtained by the lcst, due to clouds, and secondly to the MERRA-2 interpolation which contains errors that we are not considering.

5.1.1.2 Exponential

Just as before, considering the total points of each sensor, we were able to adjust the exponential curve as follows:

- For the BCC sensor:

$$PWV = 32.02 \cdot \exp\left(\frac{T_{sky}}{12.56}\right) + 1.00 \quad (5.1.6)$$

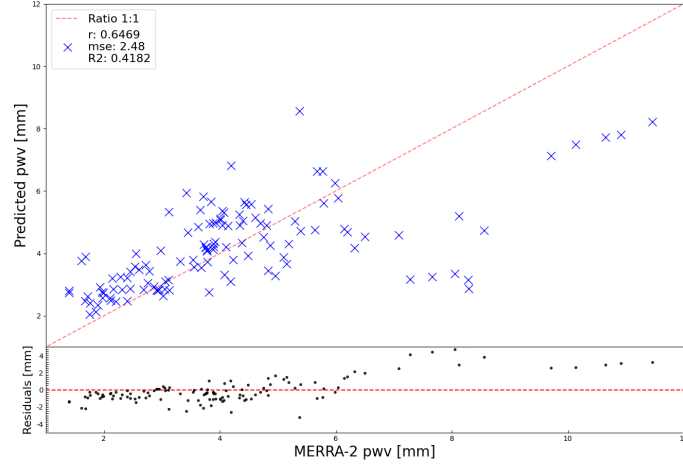


Figure 5.1.4: Comparison of the MERRA-2 pwv versus the predicted pwv. The red line is a ratio 1:1. This prediction follows the equation 5.1.6, and has mse=2.47, r=0.64 and R2=0.41.

- For the BBA1 sensor:

$$PWV = 19.27 \cdot \exp\left(\frac{T_{sky}}{23.00}\right) + 1.00 \quad (5.1.7)$$

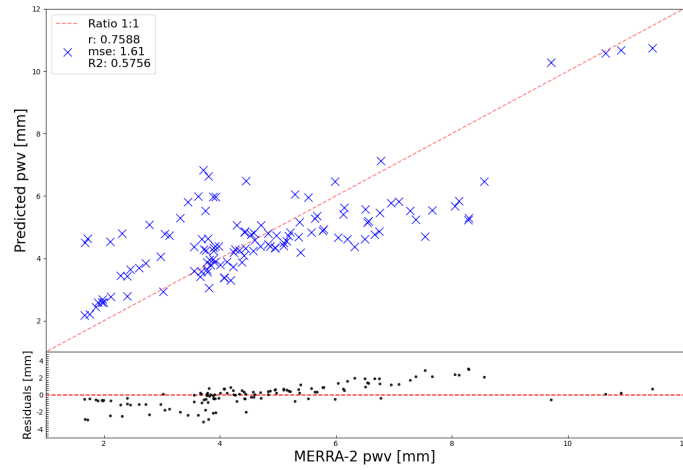


Figure 5.1.5: Comparison of the MERRA-2 pwv versus the predicted pwv. The red line is a ratio 1:1. This prediction follows the equation 5.1.7, and has mse=1.61, r=0.75 and R2=0.57.

- For the BBA2 sensor:

$$PWV = 6.80 \cdot \exp\left(\frac{T_{sky}}{48.26}\right) + 1.00 \quad (5.1.8)$$

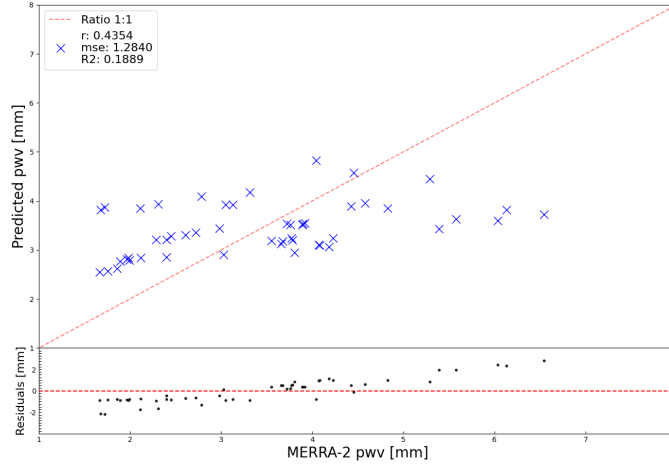


Figure 5.1.6: Comparison of the MERRA-2 pwv versus the predicted pwv. The red line is a ratio 1:1. This prediction follows the equation 5.1.8, and has mse=1.28, r=0.43 and R2=0.18.

Similar to the linear fits, we use CURVE FIT to adjust the free-parameters a , b and c . In general, this prediction are similar to the linear ones, having similars r and $R2$ in the BCC and BBA1 sensors, but differents for the BBA2. This could be due to the lack of data for this sensor, being less than a half of the other ones.

5.1.1.3 Clausius-Clapeyron

Here we can not use LCST measurements, because the instrument located at GARS does not have the BME280 sensor in the outside, so we can only make predictions of the PWV using the MeteoCHILE dataset.

We have to modified the coefficients of the equation 4.4.1, as they don't match the true pwv, so we consider them as free-parameters, and using the LEAST SQUARED function of Scipy, we found the following equation:

$$PWV = 2.7 \times 10^{-8} \cdot RH \cdot e^{\left(\frac{1 + 1.3 \cdot \frac{RH \cdot e}{p}}{T} \right)} \quad (5.1.9)$$

$$e = \exp \left(\frac{34.1 \cdot T}{T + 556.9} \right) \quad (5.1.10)$$

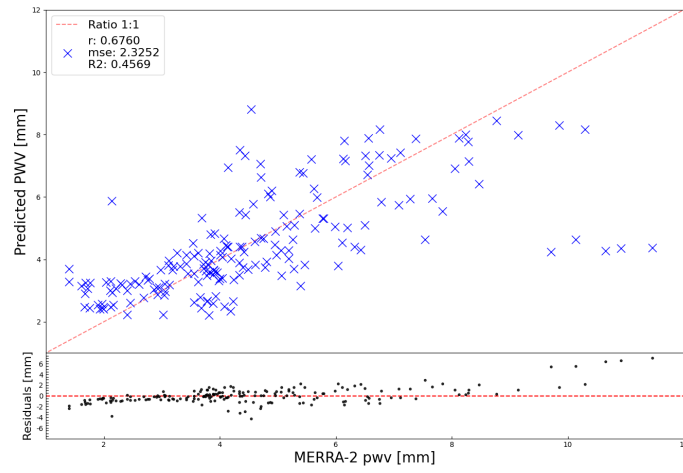


Figure 5.1.7: Comparison of the MERRA-2 pwv versus the predicted pwv. The red line is a ratio 1:1. This prediction follows the equation 5.1.9, and has mse=2.32, r=0.67 and R2=0.45.

Even though there is a trend, the scatter is greater. This is represent by the low R^2 value.

5.2 APEX, Chajnantor, Chile

In addition to the exponential relationship, we have found a quadratic behavior between the infrared sky temperature and the pwv, so we will fit another equation:

$$PWV = \frac{(T_{sky} + a)^2}{b} + c \quad (5.2.1)$$

where a , b and c are free-parameters to adjust.

5.2.1 Mathematical Relationships

For the lcst located at APEX, we will only fit the exponential relationship, according to the equation 3.3.2, and the Clausius-Clapeyron equation. Similar to the previous fits, we use CURVE FIT from Scipy to adjust a , b , c and d as free-parameters. As mentioned in 4.2, we will not consider the data when the sun passes in front of the sensors.

For this relationships, we use the more stables days, described in table 4.2.1. As we mention, we have one measurement each minute, so we have a total of 17280 points. By eliminating the data of the sun contribution, we end with 13438 data for the BCC sensor, and 9058 data for the BBA1 and BBA2.

5.2.1.1 Exponential

- For the BCC sensor:

$$PWV = 13.95 \exp\left(\frac{T_{sky} + 29.19}{9.12}\right) - 0.18 \quad (5.2.2)$$

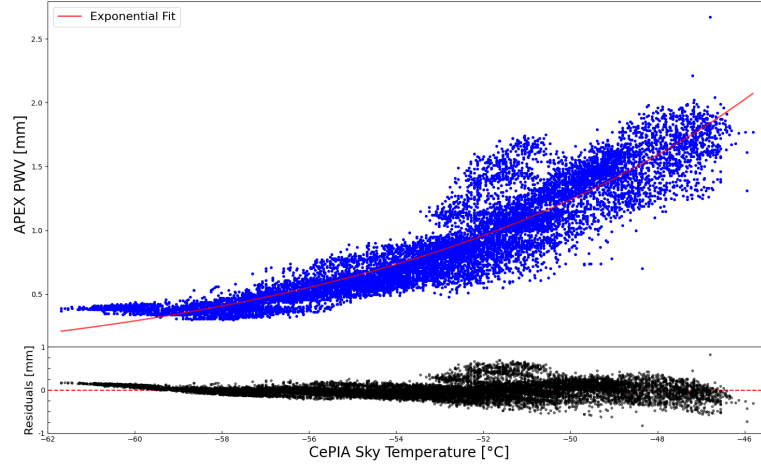


Figure 5.2.1: Infrared sky temperature versus PWV from the APEX radiometer. In red is the best exponential fit to the data. In the bottom panel are the residuals.

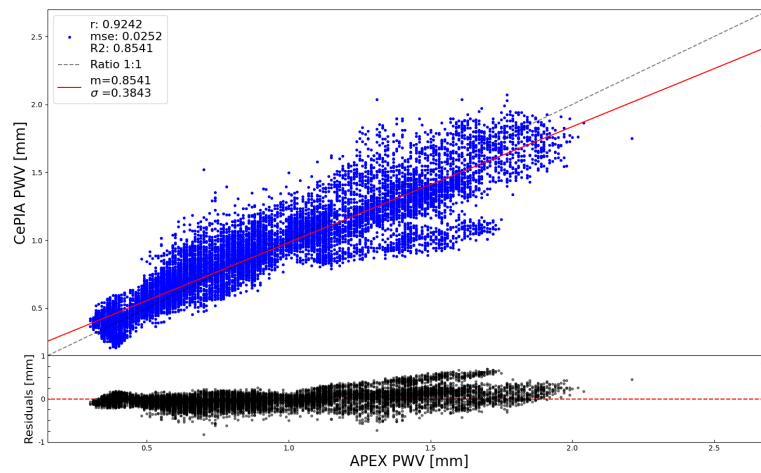


Figure 5.2.2: Comparison of the APEX pwv versus the predicted pwv. The black line is a ratio 1:1, and the red continuous line is the best fit for the data, with slope $m=0.85$. The data has dispersion $\sigma = 0.38$, $r=0.92$, $R^2=0.85$ and $mse=0.02$.

- For the BBA1 sensor:

$$PWV = 11.61 \exp\left(\frac{T_{sky} + 28.87}{10.52}\right) - 0.22 \quad (5.2.3)$$

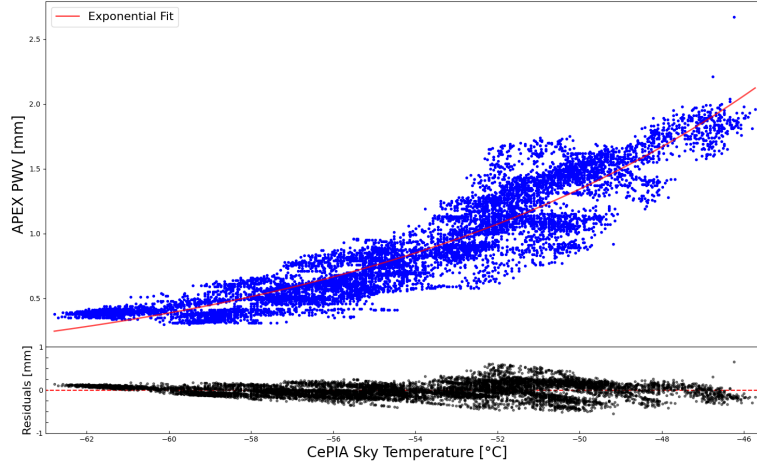


Figure 5.2.3: Infrared sky temperature versus PWV from the APEX radiometer. In red is the best exponential fit to the data. In the bottom panel are the residuals.

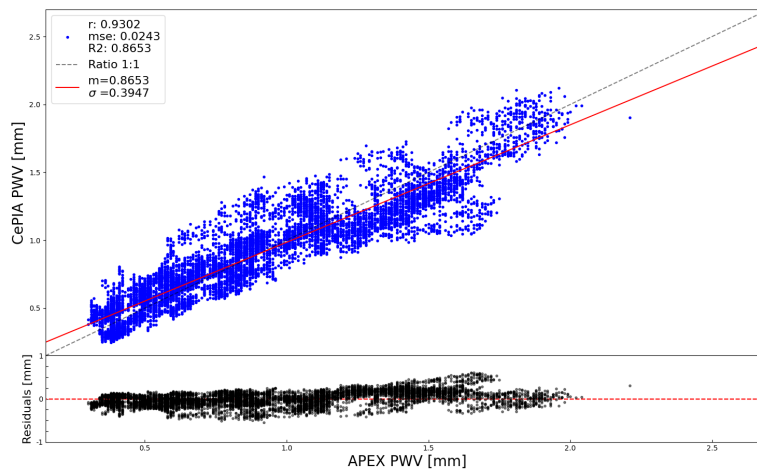


Figure 5.2.4: Comparison of the APEX pwv versus the predicted pwv. The black line is a ratio 1:1, and the red continuous line is the best fit for the data, with slope $m=0.86$. The data has dispersion $\sigma = 0.39$, $r=0.93$, $R^2=0.86$ and $mse=0.02$.

- For the BBA2 sensor:

$$PWV = 11.37 \exp\left(\frac{T_{sky} + 28.53}{10.69}\right) - 0.22 \quad (5.2.4)$$

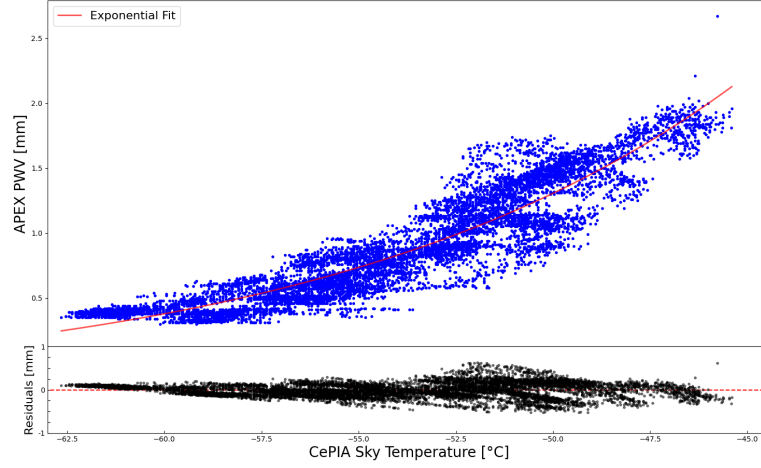


Figure 5.2.5: Infrared sky temperature versus PWV from the APEX radiometer. In red is the best exponential fit to the data. In the bottom panel are the residuals.

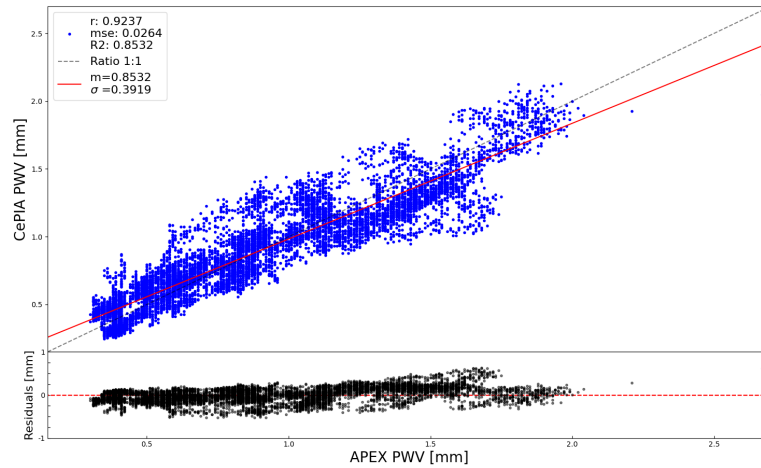


Figure 5.2.6: Comparison of the APEX pwv versus the predicted pwv. The black line is a ratio 1:1, and the red continuous line is the best fit for the data, with slope $m=0.85$. The data has dispersion $\sigma = 0.39$, $r=0.92$, $R^2=0.85$ and $mse=0.02$.

5.2.1.2 Quadratic

- For the BCC sensor:

$$PWV = \frac{(T_{sky} + 61.65)^2}{145.54} + 0.32 \quad (5.2.5)$$

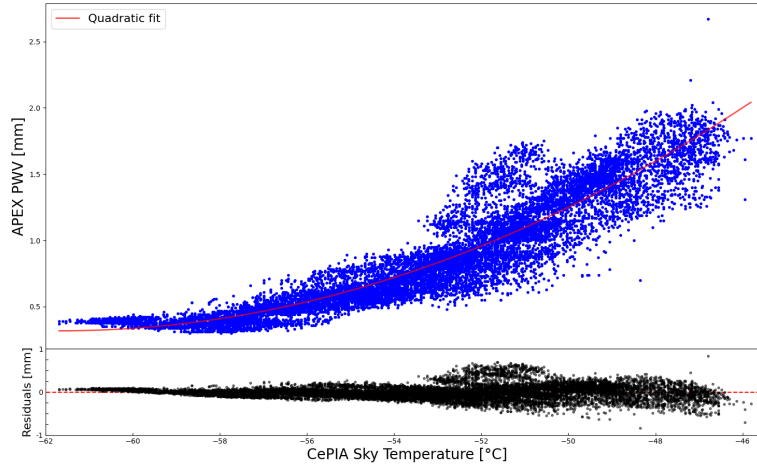


Figure 5.2.7: Infrared sky temperature versus PWV from the APEX radiometer. In red is the best quadratic fit to the data. In the bottom panel are the residuals.

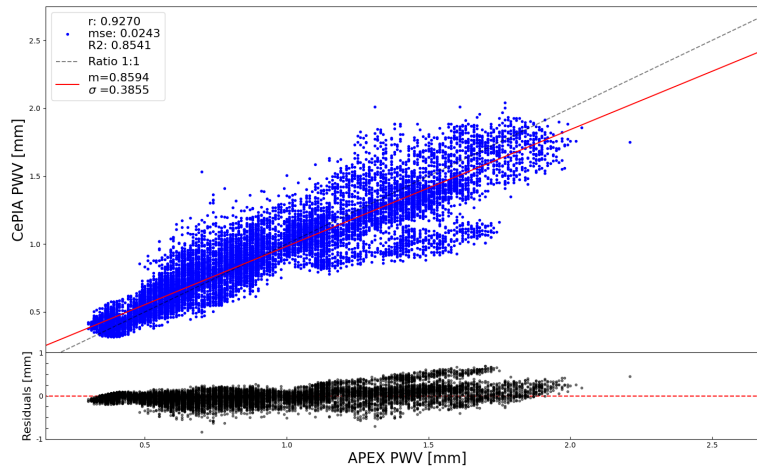


Figure 5.2.8: Comparison of the APEX pwv versus the predicted pwv. The black line is a ratio 1:1, and the red continuous line is the best fit for the data, with slope $m=0.85$. The data has dispersion $\sigma = 0.38$, $r=0.92$, $R2=0.85$ and $mse=0.02$.

- For the BBA1 sensor:

$$PWV = \frac{(T_{sky} + 63.74)^2}{182.49} + 0.32 \quad (5.2.6)$$

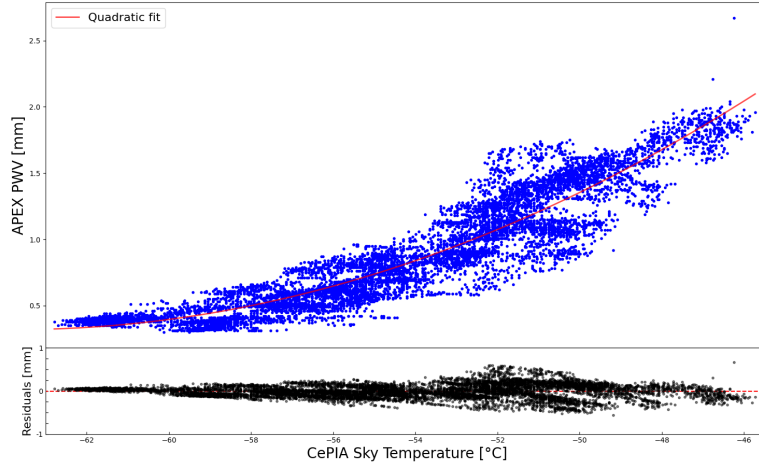


Figure 5.2.9: Infrared sky temperature versus PWV from the APEX radiometer. In red is the best quadratic fit to the data. In the bottom panel are the residuals.

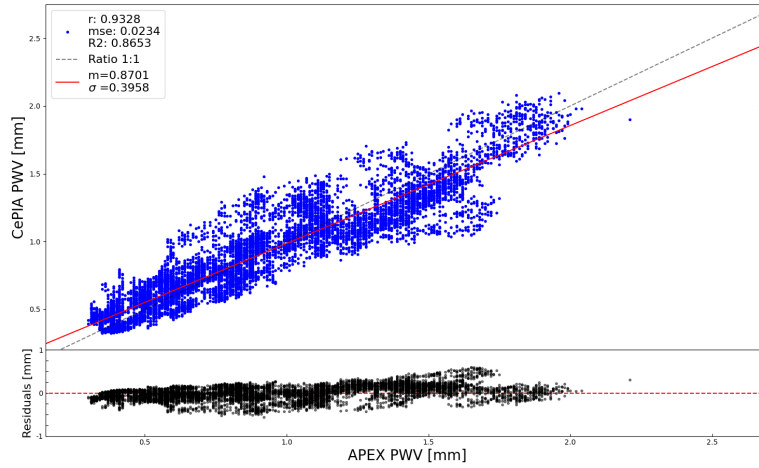


Figure 5.2.10: Comparison of the APEX pwv versus the predicted pwv. The black line is a ratio 1:1, and the red continuous line is the best fit for the data, with slope $m=0.87$. The data has dispersion $\sigma = 0.39$, $r=0.93$, $R^2=0.86$ and $mse=0.02$.

- For the BBA2 sensor:

$$PWV = \frac{(T_{sky} + 63.73)^2}{188.40} + 0.32 \quad (5.2.7)$$

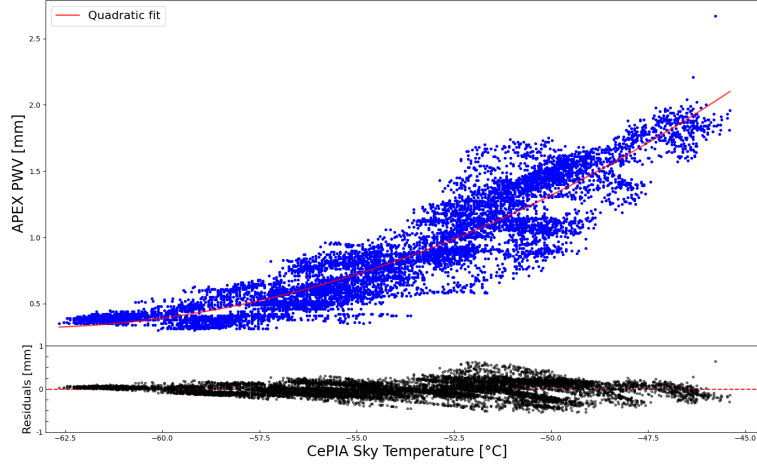


Figure 5.2.11: Infrared sky temperature versus PWV from the APEX radiometer. In red is the best quadratic fit to the data. In the bottom pannel are the residuals.

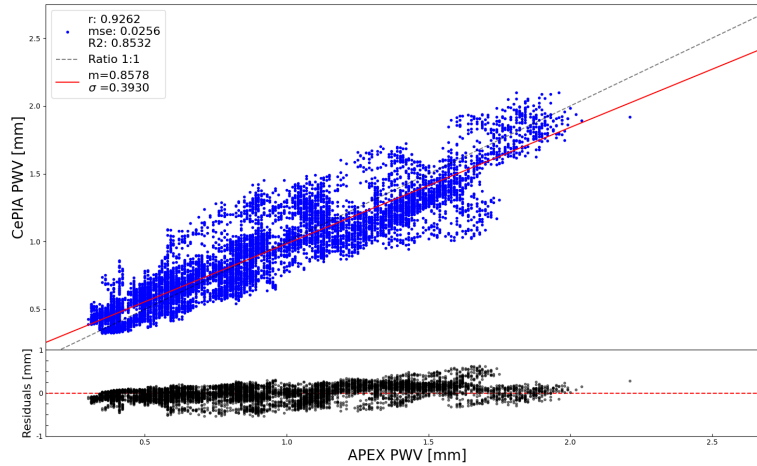


Figure 5.2.12: Comparison of the APEX pwv versus the predicted pwv. The black line is a ratio 1:1, and the red continuous line is the best fit for the data, with slope $m=0.85$. The data has dispersion $\sigma = 0.39$, $r=0.92$, $R2=0.85$ and $mse=0.02$.

It is clear that the improvement is great. In table 5.2.1 we can see a comparative between both models.

EXPONENTIAL				QUADRATIC		
Sensor	r	MSE	R2	r	MSE	R2
BCC	0.9242	0.0252	0.8541	0.9270	0.0243	0.8541
BBA1	0.9302	0.0243	0.8653	0.9328	0.0234	0.8653
BBA2	0.9237	0.0264	0.8532	0.9262	0.0256	0.8532

Table 5.2.1: Comparison of exponential and quadratic metrics

We can clearly see that the metrics get better, by obtaining a Pearson coefficient greater than 0.92, MSE less than 0.026 and R2 score greater than 0.85 in all cases. In addition to this, by fitting a simple linear regression to the predicted data, we obtain a slope greater than 0.85 in all cases, which is a good approximation.

5.2.2 BME-280 Calibration

In order to predict pwv through the Clausius-Clapeyron equation (equation 4.4.1), we need ambient temperature, specific humidity and pressure data. In this version of the instrument, the BME-280 sensor is in the outside, so we have this measurements.

We use the meteorological station of APEX to compare this measurements, so we can calibrate our instrument. The first thing to notice, is that in the lcast the pressure measurements lacks of resolution. It gives values of pressure in [kPa], with resolution of 1 [kPa], so we can not access to the same resolution that the meteorological station of APEX, which brings values of [hPa], with resolution of 0.01 [hPa], which is 1000 times better resolution.

For the ambient temperature and humidity we can work as follows.

5.2.2.1 Specific Humidity

For this, we use all the available data, and the measures are similiar to a 1:1 ratio, but with an offset. To solve this, we only have to calculate the offset, and then substract it to the data. To calculate the offset, we create a function that minimizes the mean squared error between the APEX meteorological station

humidity measurements and the lcst humidity measurements for each day. Finally, we average all of these values for all the days, and we obtain a value of 3.6842.

This correction can be seen in Figure 5.2.13.

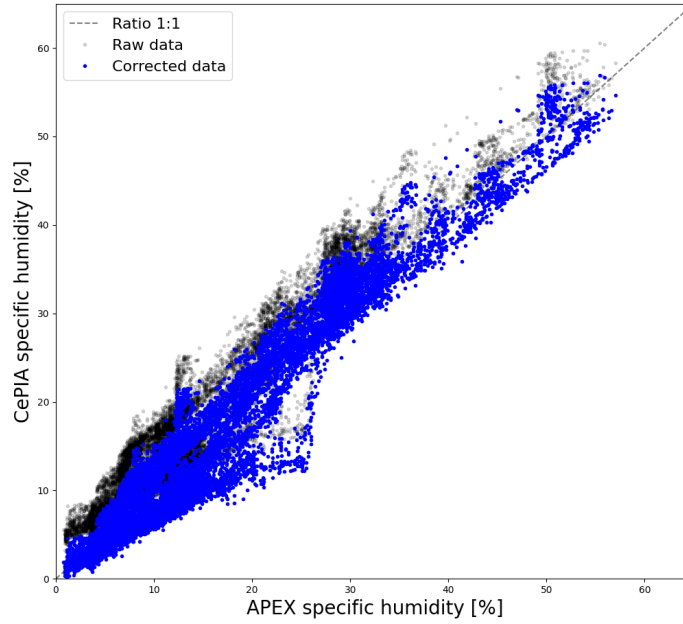


Figure 5.2.13: APEX meteorological station measurements of humidity versus de LCST humidity. In black dots we see the raw data, with an offset. In blue dots is the corrected data, calculated as $\text{Humidity} = \text{Humidity} - 3.6842$.

5.2.2.2 Ambient Temperature

For calibrating the temperature, we see that below the 0°C , the lcst sensor underestimate the true temperature in an, almost, linear relationship, and above the 0°C , the lcst sensor overestimate the true temperature, in a non-linear manner.

Using a similar method of the humidity calibration, for each day we calculate the mean value of temperature, and then we average all of these means. Then, by adjusting a simple linear fit, with the slope equal to the average of the means, $m=0.89$, and by calculating the standard deviation of the data, $\sigma = 6.64$, we can set a reliable set of temperatures, as seen in Figure 5.2.14.

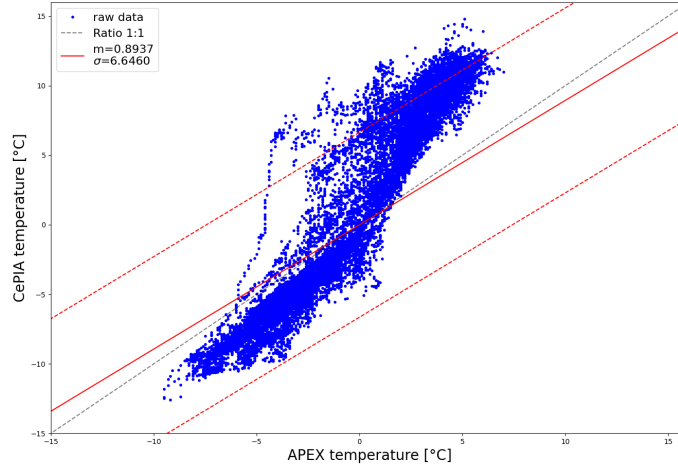


Figure 5.2.14: APEX meteorological station measurements of temperature versus de LCST temperature. In blue dots we see the raw data. In red is the average of the averages values with 1σ .

Now to this new set of data, we adjust a piecewise function, in order to normalize the data, by adjusting the following equation:

$$LCST_{tempcorr} = LCST_{temp} \cdot \frac{APEX_{temp}}{f(x)} \quad (5.2.8)$$

where $f(x)$ is a function that best fit in the data. As we have a division, we transform all the temperatures to Kelvin, in order to avoid the division by zero. Then, by adjusting several functions, the one that most fitted the data is a piecewise function defined as:

$$f(x) = \begin{cases} 1.35T_{amb} - 97.39 & , \quad x \leq 272.46 \\ -0.08T_{amb}^2 + 47.08T_{amb} - 6522.87 & , \quad 272.46 < x < 275.34 \\ 1.38T_{amb} - 98.74 & , \quad x \geq 275.34 \end{cases} \quad (5.2.9)$$

With this correction, we have the final corrected temperature as shown in Figure 5.2.15:

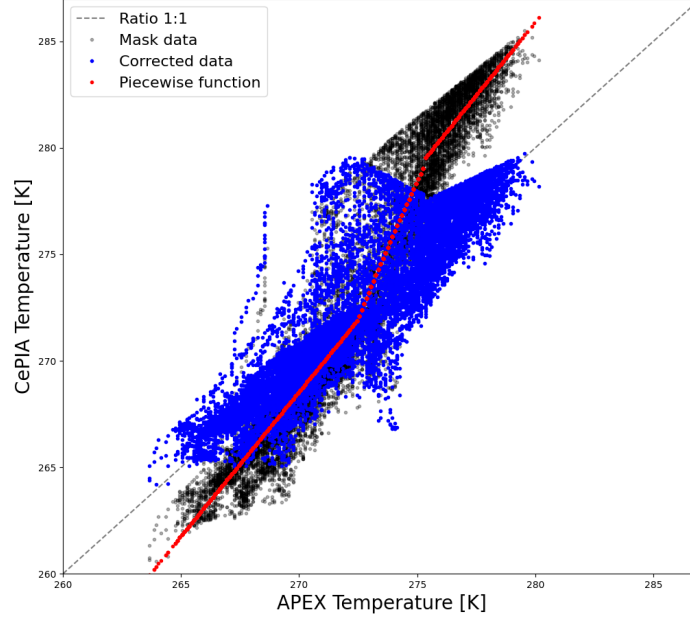


Figure 5.2.15: Comparison of APEX temperature versus LCST temperature. The dashed line is a ratio 1:1. The black dots is the set of data of 1σ (see fig 5.2.14). The red dots are the piecewise function. The blue dots are the corrected temperature data, by following the equation 5.2.9.

With these corrections, we could predict the PWV by using the Clausius-Clapeyron equation. However, we need to cover the pressure values, so even if we use the APEX pressure measurements, we can not apply the Clausius-Clapeyron equation, due that at higher altitudes, like APEX >5000 masl, the atmosphere does not act the same as at sea level; the dry components of the atmosphere making that the normal conditions of pressure and temperature does not apply, which is an assumption to the equation.

Chapter 6

Conclusion

We have conducted an exhaustive work on the calibration of the Low-Cost Site Testing Instrument to characterize the atmosphere developed by CePIA. We have studied two different versions of this instrument, one located at the Antarctic Plateau, and the second one located near the APEX telescope, in the Chajnantor Plateau.

We have found several equations that will allow us to predict the Precipitable Water Vapor from the Infrared Sky Temperature: a linear relationship, an exponential relationship, and a quadratic relationship. This relationship depends roughly on the sky temperature, and it can help us to reduce the costs at the moment of site testing a possible location for the installation of radiotelescopes.

On the objective of this project, we can enumerate some of the results:

On the Antarctic study:

- We make predictions of the Sky Temperature by considering the ambient temperature and the PWV over the GARS location, at Antarctic. This yields on poor metrics, with Pearson coefficient lower than 0.8 in all of the cases.
- We found an exponential relationship by considering only the sky temperature and we predict the PWV over the GARS location, at Antarctic. This also yields on poor metrics, with Pearson coefficient lower than 0.8 in all of the cases as well.

- Moreover, we try to predict the PWV by using a modified Clausius-Clapeyron equation, which depends on the ambient temperature, the specific humidity and the pressure, but this prediction also have poor metrics, with a Pearson coefficient less than 0.7.
- In conclusion, the LCST located at GARS could not provide good results for the global aim. This is due the extreme weather conditions in Antarctic, where out of 9 months, we could only rescue less than 2 months of <usefull> data, which we can only trust based on the cloud cover provided by MeteoCHILE, in oktas.

On the APEX study:

- We have found a clear trend on how the pwv behaves with respect to the infrared sky temperature.
- We adjust two different fits to the data: an exponential and a quadratic equation. Both of which produced good results.
- The exponential fit yield in a Pearson coefficient greater than 0.92 in all of the cases, same as the quadratic equation.
- Furthermore, we calibrated an ambient sensor, by adjusting both the humidity sensor and the ambient temperature sensor.
- The humidity sensor only needed an offset adjust, while the temperature sensor needed a piecewise function to be corrected.

With these results, we can conclude that the Low-Cost Site Testing Instrument could correctly reproduce the Precipitable Water Vapoer measurements by only measuring the infrared sky temperature, in clear-sky conditions.

The errors associated to each temperature measure is still on study.

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